

Measuring the gap between declared and real urgency

SupplyScore, a cognitive digital twin that tells you when you are **not worried enough**, before the line stops.

John HOARAU 484338 | R&D Intern, Direction Innovation Numérique, ALTEN | Individual Research Project 2025/2026

THE THREAD Every honest measurement in this project **contradicted a belief**, and each contradiction is why the instrument can now be trusted.

ONE HYPOTHESIS SURVIVED, AND IT IS THE INTERESTING ONE

Six falsifiable hypotheses were frozen before any data existed. Five were refuted by their own thresholds. The survivor: **urgency is contagious**. Nodes with no physical disruption still raised their declared urgency by **+0.079** on press-event turns, against **+0.005** on calm turns.

HAS, LLM pilot, 8 nodes × 19 turns.

THE MODEL RANKS. IT DOES NOT CALIBRATE.

Scored against a frozen ground truth, the forecast layer discriminates (**AUC 0.736** at one turn) yet its Brier skill is **negative at every horizon**. Below 10 % it understates the real rate **3.3x**; above 90 % it overstates it **4x**. Usable for triage, not yet for a number a manager acts on.

120 prospective forecasts, turns 4–18.

THE WRONG TARGET NEARLY BURIED IT

Same model, same predictions, same data, four-turn horizon: **AUC 0.435** when scored against “any event on the node”, **AUC 0.654** against the frozen definition of an adverse outcome. **An evaluation against the wrong target can condemn a system that works.**

Caught only because the definition was frozen in one place.

1 THE BLIND SPOT Everything is green. The line stopped anyway.

- Two failures hide in every dashboard, and they have **opposite signs**
- False urgency** $F = [U_d - U_r]_+$: everyone cries wolf, premium capacity saturates, the next alert is believed a little less
- Hidden risk** $H = [U_r - U_d]_+$: the data screams, nobody declares, the rupture arrives with the options already expired
- Panic is expensive and recoverable. Silence is cheap and irreversible
- So the score is **deliberately asymmetric**: same divergence of 0.5 scores $A = 53.1$ for panic and $A = 21.1$ for silence

The dashboard was not wrong. It was measuring the wrong thing.

2 THE GAP IN THE FIELD Everyone scores the world. Nobody scores the gap.

- Ripple-effect models, Markov and dynamic Bayesian disruption models, Bayesian supply-chain risk networks, TTR/TTS stress tests
- Commercially: Everstream, Resilinc, Interos, mapping multi-tier networks at a scale no research prototype reaches
- All of them score **physical reality**. None records the human declaration as a quantity, so none can subtract one from the other
- Not claimed as novel**: the propagation algebra (a classical survival product), AHP, PROMETHEE II, Monte Carlo PERT, noisy-OR. Every method here is established
- Claimed**: the quantity they are assembled to compute, and the discipline imposed on the assembly

Everyone scores the world. Nobody scores the gap between the world and what we say about it.

3 WHAT WAS BUILT Two flows, one score

- U_r , **real urgency**: six KPI blocks (time, capacity, performance, risk, cost, CO₂) combined by weighted **probabilistic OR**, not a mean
- Why: a total material rupture with five healthy blocks **averages to 0.17** while the line is physically stopped. One saturated detector must trip the alarm
- A missing block is **ignored, never imputed**: absence of measurement is not evidence, in either direction
- U_d , **declared urgency**: weekly AHP over four fixed criteria, with a blocking consistency gate at CR < 0.10. Block weights by Fuzzy Best-Worst Method
- Propagation**: declared need descends the DAG attenuated by γ , physical risk climbs it scaled by β
- A(t)**: Prospect-Theory asymmetric loss, $\lambda_{\text{under}} = 2.25$, $\alpha = 0.88$, both measured by Tversky and Kahneman, not tuned here

We changed the mathematics because the arithmetic disagreed with the factory floor.

4 DELIVERED, AND AUDITABLE A score that can be argued with

- 86 Python modules** across 13 packages, **1 974 automated tests**, coverage gate blocking below 90 %, thirteen application pages
- One write path**. Every business write is validated, diffed and audited inside the same transaction as the change. The local CI greps the UI layer for direct writes and **fails the build** if it finds one
- Exact decompositions, not attributions**. U_d splits additively by criterion; U_r splits exactly in log-survival space by block
- So a player who contests a score of 46.5 is answered **term by term**, on screen, with the audit trail underneath
- Append-only history, WAL databases, integrity-checked backups, three clocks (real, game, fixed)

The score's job is not to be believed. It is to be checkable.

5 THE TEST HÉLIOS: the chip crisis, replayed blind

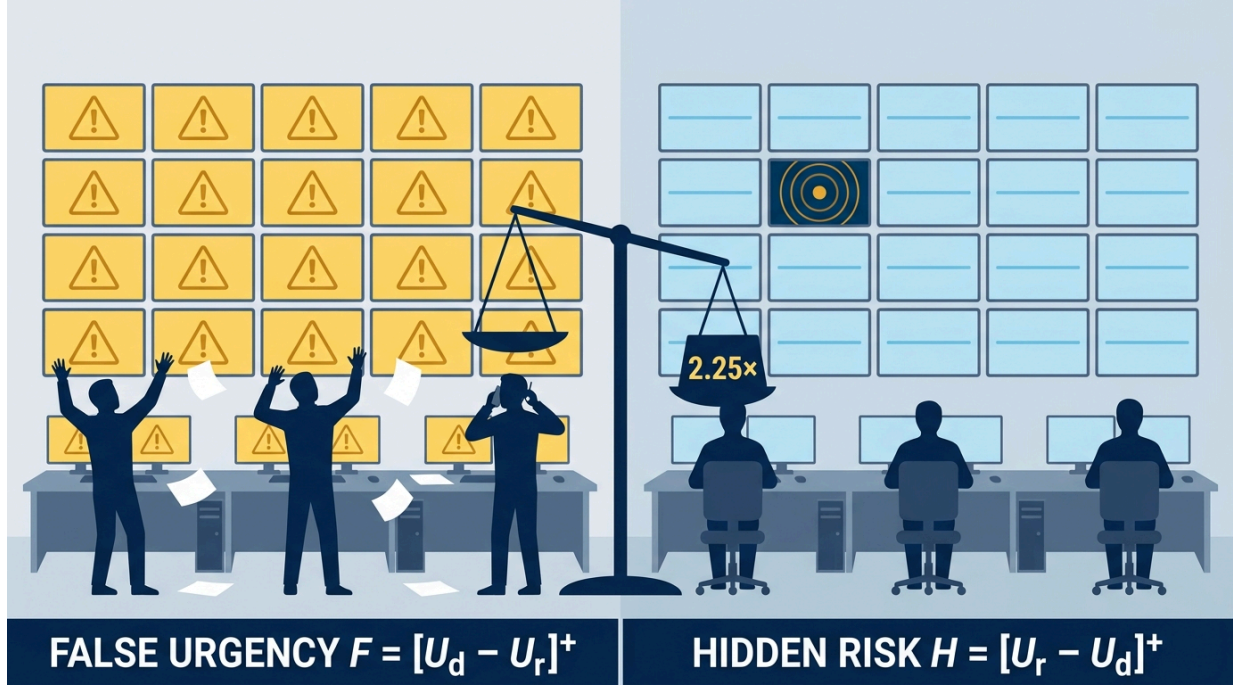
- The 2020–2022 semiconductor shortage rebuilt as an **8-node, 18-turn** serious game under narrative cover. Players are never told which crisis they are in
- U_r fed by authentic public series (INSEE, WSTS, BLS) frozen under SHA-256; twelve calibrated events pinned to dated real ones: Texas freeze, Renesas fire, Suez, Taiwan drought, Malaysian back end, the autumn bullwhip
- Pre-registered and frozen at day five**: six hypotheses, numeric thresholds, and the analysis code all written **before any data existed**
- A **null baseline** imposed on itself (precision 0.16, recall 0.27), structural anti-lookahead in the briefing generator, two placebo briefings, and one third-party dataset **rejected** at the authenticity check

We wrote down how we could be wrong before we could know whether we were.

6 WHAT BROKE, AND WHAT IS NEXT The instrument found four faults in itself

- Milestone risk is blind to events**. It reacts only to nominal lead time, and no calibrated event routes there. At turn 6 it announced **99.6 %** on a milestone delivered on time and **0.0 %** on one that was missed while that factory sat frozen. Near anti-correlated
- The shock did not propagate**, upstream or downstream, on a model whose reason to exist is propagation
- Calibration fails in both directions at once**, which is why the mean looks almost right while the skill score is deeply negative
- The ladder was climbed out of order**: the historical replay ran before the model had ever been trained. That mistake prevented a costlier one, since running the synthetic factory first would have taught a model to be well calibrated on noise
- Next, in this order**: repair the core → run the factory → train and calibrate → re-test HÉLIOS against the archived before-repair measurements → the human campaign

A model that is confident where it is blind is worse than no model.

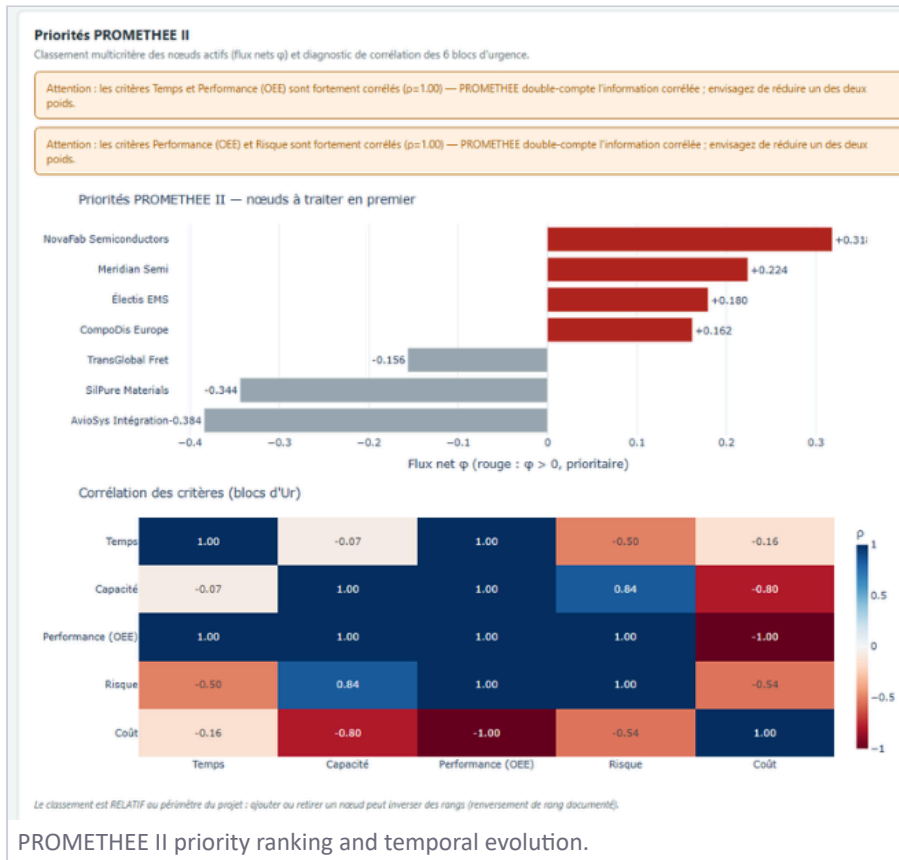
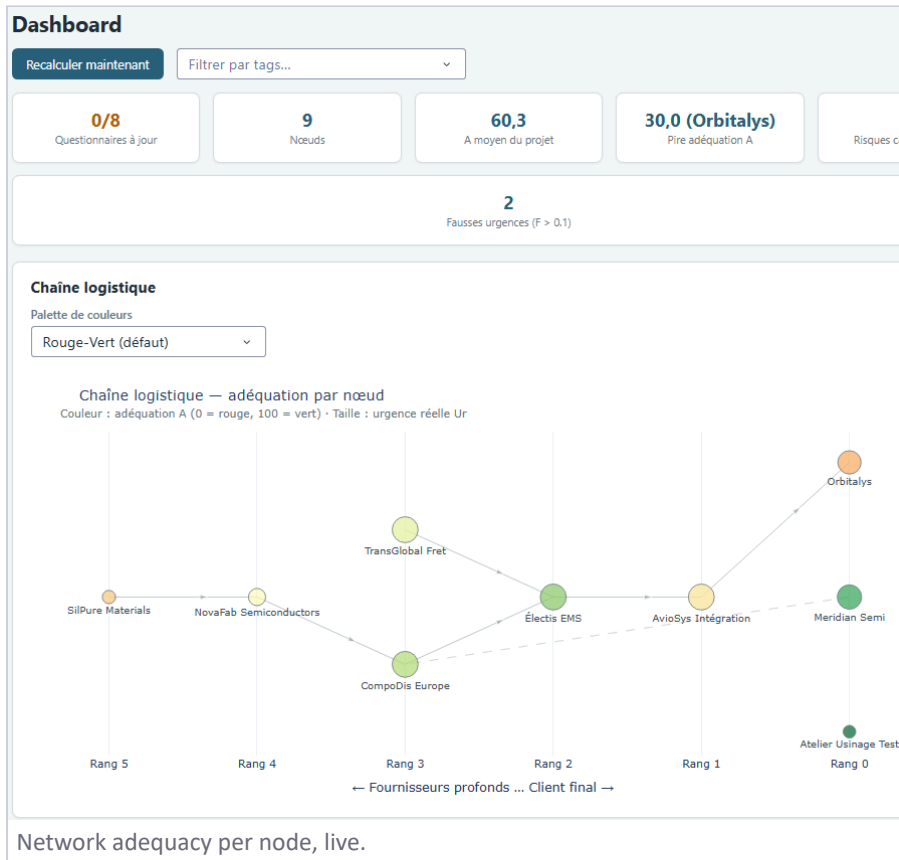
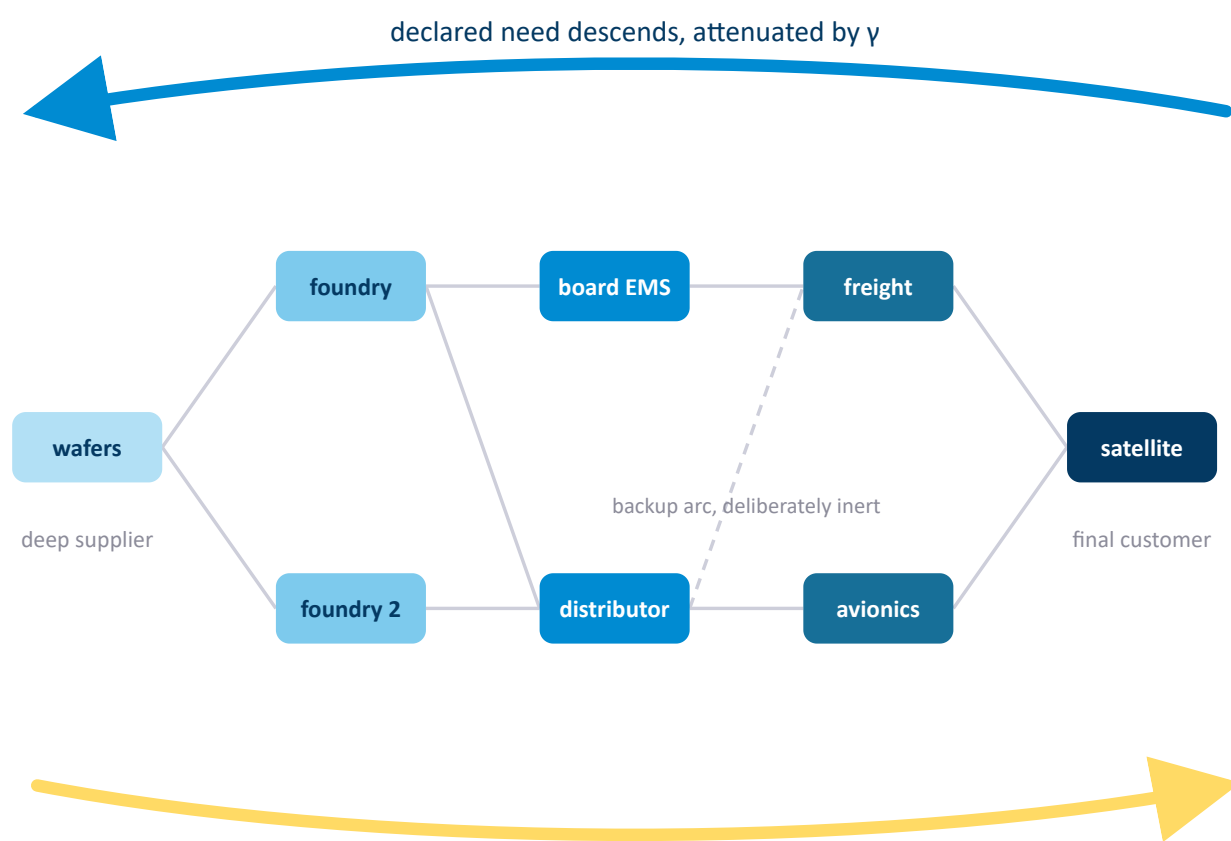


	Ripple-effect models	Markov and dyn. Bayesian	Bayesian SC risk networks	TTR / TTS stress tests	Everstream, Resilinc, Interos	SupplyScore
Scores physical reality	✓	✓	✓	✓	✓	✓
Scores human declaration	✗	✗	✗	✗	✗	✓
Scores the gap	✗	✗	✗	✗	✗	✓



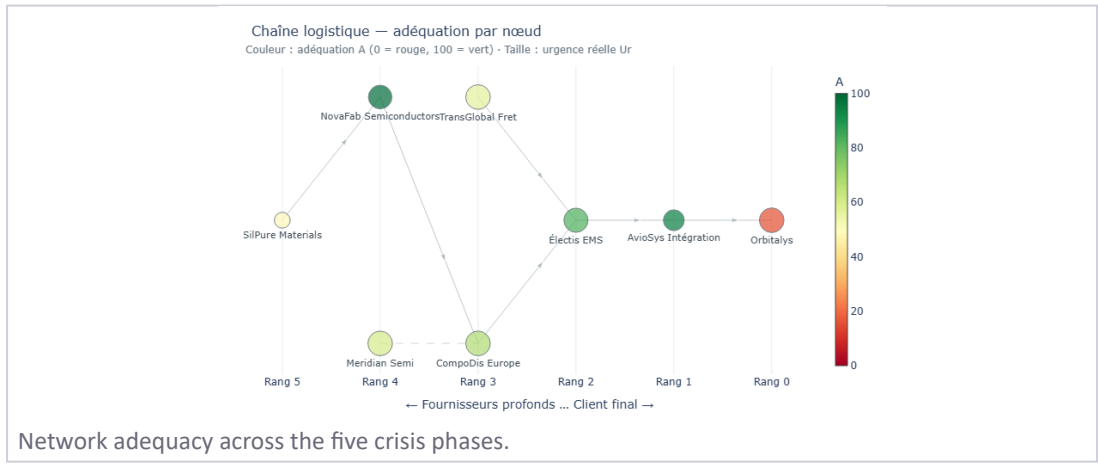
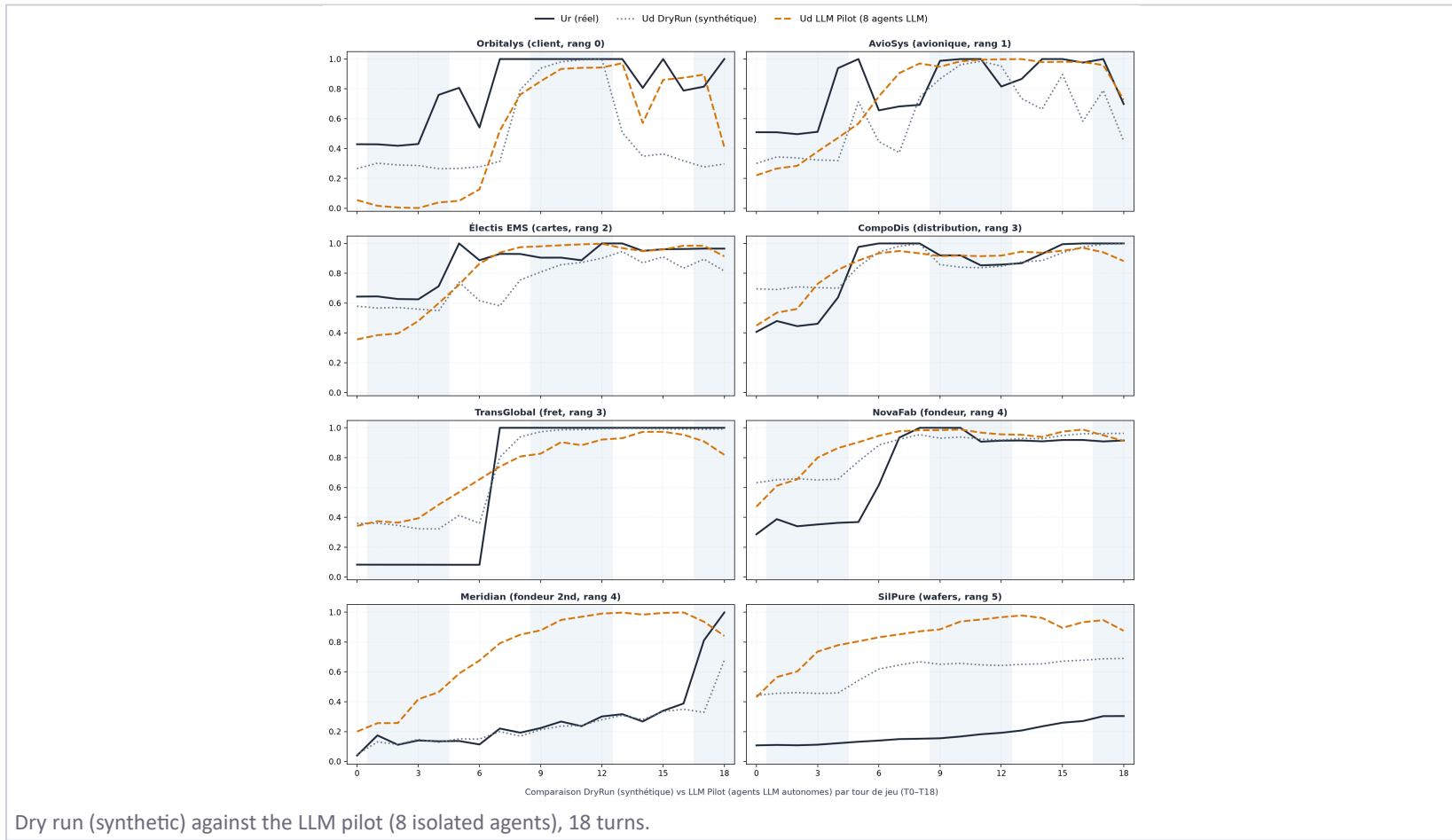
$$U_{r,local} = 1 - \prod_m (1 - u_m)^{\omega_m}$$

One saturated block is enough. A mean would have buried it.



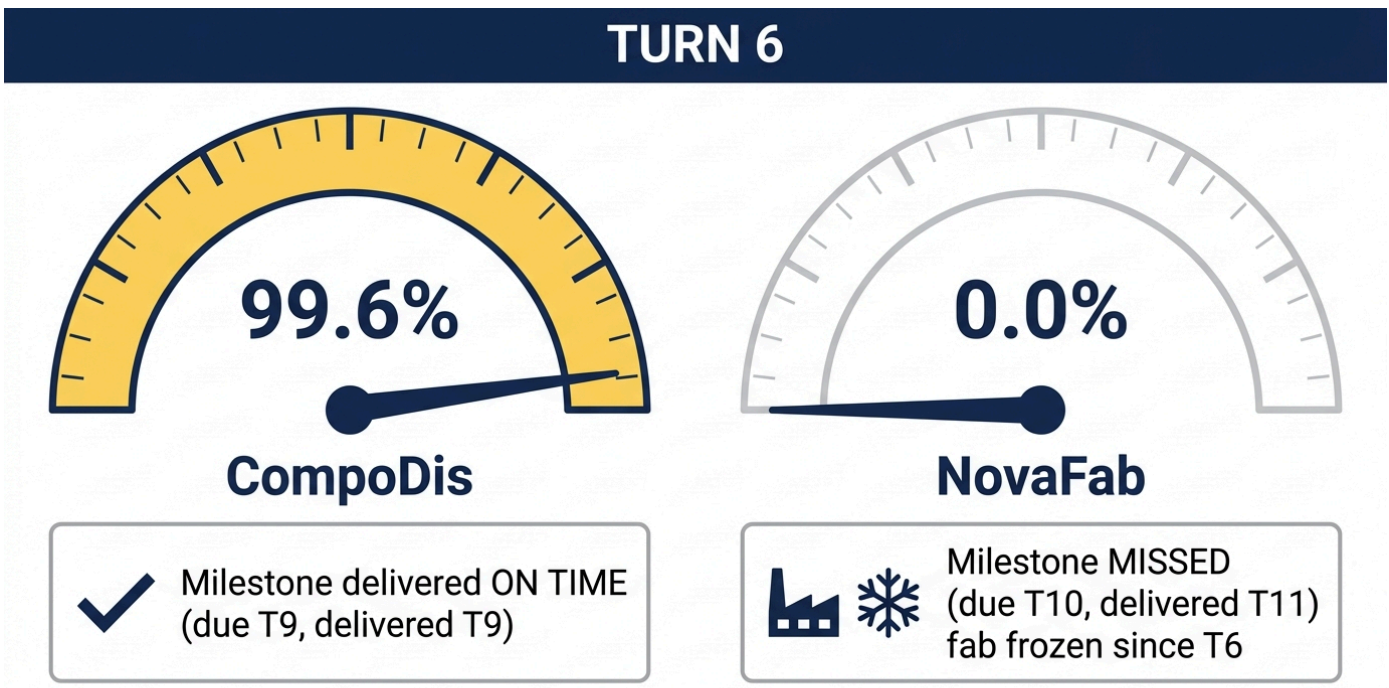
Détail des nœuds						
Fond rouge clair : A < 40 - H en rouge : H > 0.3 - Hebdoo coloré selon le statut (A/F/H « — » si questionnaire manquant)						
Nom	Rang	Statut	Hebdoo	Ud_loc	Ur_loc	Ud
filter data...						
Atelier Usage Test	0	Active	Manquant	0	0	
Meridian Semi	0	Active	En retard (1 sem)	0.841	0.879	0.841
Orbitalys	0	Terminée	En retard (1 sem)	0.409	0	0.409
AvicSys Integration	1	Active	En retard (1 sem)	0.568	1	0.727
Électis EMS	2	Active	En retard (1 sem)	0.793	1	0.914
CompoDis Europe	3	Active	En retard (1 sem)	0.561	1	0.882
TransGlobal Fret	3	Active	En retard (1 sem)	0.666	1	0.819
Novafab Semiconductors	4	Active	En retard (1 sem)	0.575	0.325	0.912
SiPure Materials	5	Active	En retard (1 sem)	0.656	0.048	0.876

Per-node scores: U_d , U_r , A, F, H.



Network adequacy across the five crisis phases.

Pre-registered hypothesis	Verdict	Measured
H1 Cognitive latency	Not confirmed	3 of 8 nodes (bar 5)
H2 Early detection	Not confirmed	H max 0.017 (bar 0.3)
H3 Structural criticality	Not confirmed	top-1 on 70 % (bar 80 %)
H4 Predictive validity	Not confirmed	precision / recall 0.00
H5 Urgency contagion	Confirmed	+0.079 vs +0.005
H6 Perception hysteresis	Not applicable	no measurable decline



NEAR ANTI-CORRELATED

T1 ✓ Functioning on synthetic data. 19-turn replay end to end.	T2 — Generalisation. Skipped . The synthetic factory has never run at scale.	T3 ! Historical replay. Attempted early ; returned four faults, not validation.	T4 ☹ Utility. The causal battery fails its own validation and the pipeline withholds the artefact .
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The prescriptive layer is gated shut by the system itself: on a failed causal validation the effects artefact is never published, while the forecast still is. Nobody overrode it. The last measurement that disagreed with us was made by the system.